Status	Finished
Started	Thursday, 9 October 2025, 1:29 PM
Completed	Thursday, 9 October 2025, 1:58 PM
Duration	28 mins 5 secs
Marks	3.00/3.00
Grade	**3.00** out of 3.00 (**100**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

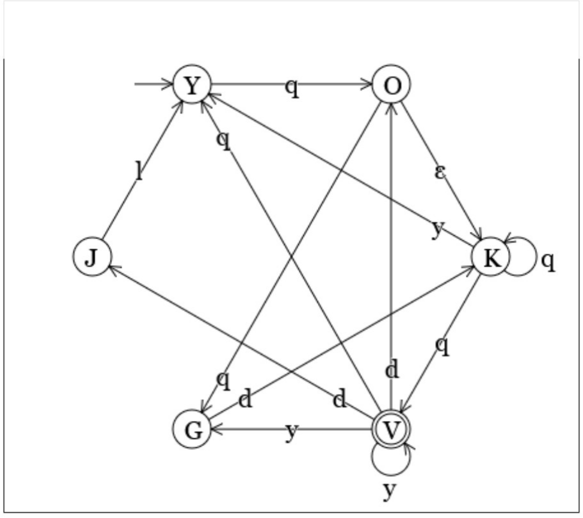
- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Translate the following DFA to a right-regular grammar.



Write the answer as a 4-tuple, separating the rule per non-terminal. Use the names of the states for their respective nonterminals.

Your answers must be syntactically correct for marking.

Terminals = {"l", "q", "y", "d"}

Your last answer was interpreted as follows:

$$\{\mathrm{l}, \mathrm{q}, \mathrm{y}, \mathrm{d}\}$$

Nonterminals = {"J","G","V","K","O","Y"}

Your last answer was interpreted as follows:

$$\{J, G, V, K, O, Y\}$$

Start symbol = "Y"

Your last answer was interpreted as follows:

$$Y$$

Rules for K = {["K", ["q", "V"]], ["K", ["q", "K"]], ["K", ["y", "Y"]]}

Your last answer was interpreted as follows:

$$\{[K, [q, V]], [K, [q, K]], [K, [y, Y]]\}$$

Rules for V = {["V", ["d", "J"]], ["V", ["y", "G"]], ["V", []], ["V", ["y", "V"]], ["V", [

Your last answer was interpreted as follows:

$$\{[V, [d, J]], [V, [y, G]], [V, []], [V, [y, V]], [V, [q, Y]], [V, [d, O]]\}$$

Rules for G = {["G", ["d", "K"]]}

Your last answer was interpreted as follows:

$$\{[G, [d, K]]\}$$

Rules for J = {["J", ["l", "Y"]]}

Your last answer was interpreted as follows:

$$\{[J, [l, Y]]\}$$

Rules for Y = {["Y", ["q", "O"]]}

Your last answer was interpreted as follows:

$$\{[Y, [q, O]]\}$$

Rules for O = {["O", ["K"]], ["O", ["q", "G"]]}

Your last answer was interpreted as follows:

$$\{[O, [K]], [O, [q, G]]\}$$

Question **2**

Correct

Mark 1.00 out of 1.00

Apply the refinement rule "Introduce Constant" (see below) to the following specification statement in order to introduce a constant R of type Nat with the value 7.

Specification statement: { g >= w } g,w:=g',w' | w' >= 17

Refinement rule Introduce Constant (InC)

{pre} v,x:=v',x' | rel

[=

CON C:T | C=x . {C=x and pre} v,x:=v',x' | rel

Fill in the correctly refined specification statement. You can copy parts of the original statement.

CON R:Nat | R=7 . { R=7 and g >= w } g,w:=g',w' | w' >= 17

Your last answer was interpreted as follows:

CON R:Nat | R=7 . { R=7 and g >= w } g,w:=g',w' | w' >= 17

Question **3**

Correct

Mark 1.00 out of 1.00

Given the set $[1, 2, 3, 4, 5]$, and relations {$[4, 3]$, $[4, 4]$, $[2, 5]$, $[1, 3]$}.

1. Fill in the relations matrix below such that all given relations are marked by a "**1**" in the correct position.
2. Then extend the set of relations by adding mandatory "**1**"'s so that the final set of relations is an equivalence relation.
3. All other entries in the matrix must be marked "**0**".

-	1	2	3	4	5
1	1	0	1	1	0
2	0	1	0	0	1
3	1	0	1	1	0
4	1	0	1	1	0
5	0	1	0	0	1

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 144

Signing code: 26581

Your last answer was interpreted as follows:

26581

◄ Test 1 (topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars)

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